

MSOE High Powered Rocketry Club

Leadership · President · Status: Ongoing

Overview

I'm President of the MSOE High Powered Rocketry Club, formerly serving as the Project Manager for the Two-Stage Rocket Team, the club's competition rocket for the 2026 Argonia Cup. I devote about 10 hours a week to the role, spanning September 2025 to present.

As President:

- Set the club's direction and oversee its project teams, fundraising, events, and certification programming.
- Organize the club's annual internal competition and the Level 1 Build Day.
- Work with our local Tripoli Rocketry Association chapter to get members through their high-power certification flights.
- Recruit and mentor new members, and keep club launches safe and on schedule.

As Two-Stage Rocket Team Project Manager:

- Lead system integration across the structural, avionics, and payload sub-teams to keep the testbed vehicle fully functional end to end.
- Design and run safety and reliability tests for the staging and dual-deployment recovery systems.
- Perform aerodynamic and physics simulations in MATLAB and OpenRocket to validate stability and predicted flight performance before committing to a build.
- Set the testbed's mission requirements: a successful stage, ignition, and safe dual-deployment recovery from roughly 3,000 ft.

Result

At the 2026 Argonia Rocket Cup, the rocket flew a fully nominal two-stage flight, staging cleanly, igniting the sustainer, and recovering under dual deployment. It reached about 14,000 ft and placed 6th of 21 teams, the club's best finish yet.

Club Events & Certification Programming

Beyond the Two-Stage Team, I help plan and run the club's annual internal competition and certification programming.

Internal Competition

Each year I help organize and run a two-tier internal competition open to students across the university, giving newcomers a structured way to design, build, and fly their own rockets.

- Level 1 (low power): teams design and build an egg-recovery rocket from club-supplied parts (motor, body tube, plywood, nosecone, parachutes) and submit an OpenRocket model plus a 3D nosecone model for board review before receiving parts. Rockets are judged in three categories: Design, Highest Launch, and an Egg Challenge (lightest rocket to recover an egg undamaged), with an overall winner scored across all three.
- Level 2 (Mid power): teams design a higher-power rocket to a club-supplied motor and body tube, carrying at least one golf ball, and must clear a minimum 700 ft apogee with a safe recovery. Scoring is entirely by flight performance calculated only for rockets that fly safely, hit minimum altitude, and stay fully intact through recovery.

Level 1 Build Day

I lead the planning and execution of our Level 1 Build Day, where members construct their own high-power rocket to fly on their Tripoli Level 1 certification flight. As part of planning it, I coordinate with the leadership of our local Tripoli Rocketry Association chapter and invite them to help lead the day, so members getting certified learn directly from experienced, certified high-power flyers, not just club leadership.

Result

The Level 1 Build Day has run for three consecutive years and helped more than 20 members earn their high-power certification. The internal competition runs every year and keeps drawing new students into hands-on rocketry. I hold Tripoli certification at Level 1 and Level 2.